

Wednesday Reading Assessment: Unit 3, Magnetic Forces and Fields

Prof. Jordan C. Hanson

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1 Memory Bank

- $V_H = Blv$... The Hall voltage, given external B-field, height of conductor, l , and the drift velocity v .
- $\vec{F} = I\vec{L} \times \vec{B}$... The Lorentz force on a current.
- $\vec{F} = I\vec{L} \times \vec{B}$... Lorentz Force on a Current
- $\vec{F} = q\vec{v} \times \vec{B}$... Lorentz Force on a Charge
- $\vec{B} = (\mu_0 I)/(2\pi r)\hat{\phi}$... The magnetic field B caused by the current I a distance r away. The field *circles* the wire, so the direction of $\vec{B}$ is $\hat{\phi}$.
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$.
- $\vec{B} = \mu_0 n I \hat{z}$... The B-field inside a solenoid. $n = N/L$, the turns per unit length.

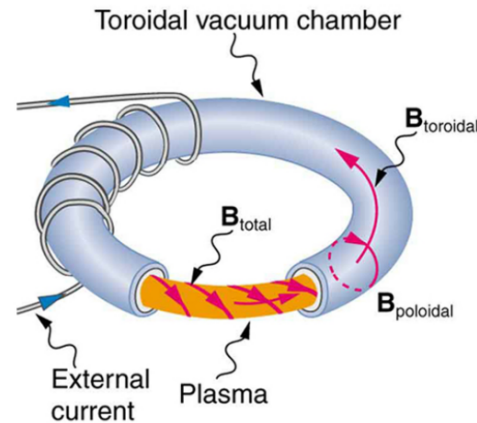


Figure 2:

2 The Hall Effect

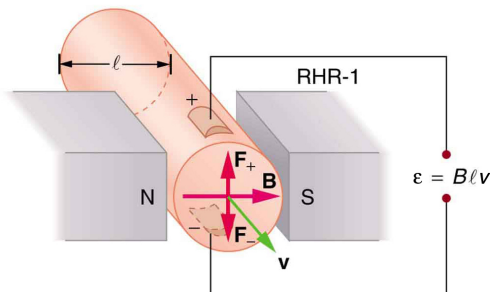


Figure 1: Measuring the Hall voltage.

1. A Hall probe is used to measure fluid flow (Fig. 1). Suppose a tube with diameter 1 cm carries a positively charged fluid moving at velocity v , in a 0.8 T B-field. The Hall voltage is measured to be $0.5 \times 10^{-6} \text{ V}$. What is the fluid velocity?
2. What would happen to the Hall voltage if the charge carriers in the fluid were *negative* instead of *positive*?

3 Ampère's Law

1. (a) What is the B-field strength 0.75 cm away from a wire carrying 1.33 A of current? (b) Describe the shape or direction of this field.
2. Consider the *tokamak* in Fig. 2. Using the right-hand rule, convince yourself that the external current creates a B -field inside the toroid that runs along the circumference of the toroid. If there are charged particles moving around the ring, trapped by the toroidal field, use the right-hand rule again to convince yourself these charged particles create a *poloidal* field.
3. Inside a *solenoid* (a linear toroid), the B-field is constant, $\mu_0 n I$, where n is the number of wire coils per unit length. (a) If we measure B to be 150 gauss inside a solenoid running 0.5 A of current, what will the B-field be at 5 A? (b) If we cut the solenoid in half, so that it is half as long, what will the B-field be at 5 A?